

Unit I :: Frames of Reference

 **PHY01MX01R04: Mechanics and Properties of Matter** 

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Syllabus

Unit I: Frames of Reference

Review of Newton's laws of motion; inertial frames of reference, Galilean transformations, Galilean invariance; non-inertial frames of reference, fictitious (inertial) forces – centrifugal force, Coriolis force; transformation equations in inclined and rotating frames.

1 Review of Newton's laws of motion

Before delving into a review of Newton's laws of motion, we need to be clear of the definitions of some important terms used in mechanics:

- *Particle*: an idealized object that has mass and position but occupies zero volume.
- *Event*: a specific occurrence that happens at a unique place in space and a specific instant in time.
- *Observer*: a person or an equipment which can locate, record, measure and interpret an event.
- *Frame of reference*: a coordinate system combined with a clock, used by an observer to measure the position and motion of particles.
- *Rest*: a particle is said to be at rest when its position does not change relative to a chosen frame of reference over time.
- *Motion*: the change in position or orientation of an object with respect to a chosen frame of reference over time.
- *Inertia*: the tendency of a body to resist any change in its motion or configuration.

1.1 Newton's first law of motion

A particle tends to maintain its state of rest or uniform motion unless it is acted upon by an external agency.

- The importance of Newton's first law of motion lies in that it defines force as:
Force is any external agency that brings about a change in the state of rest or uniform motion of a particle.

1.2 Newton's second law of motion

The rate of change of momentum of a particle is directly proportional to the force exerted on it, and takes place in the direction of the force.

$$\mathbf{F} \propto \frac{d\mathbf{p}}{dt}$$

- The importance of Newton's second law of motion lies in that it measures force as:
If a force of magnitude F brings about an acceleration a on a particle of mass m , then the force is measured to be $F = ma$.

1.3 Newton's third law of motion

For every action, there is an equal and opposite reaction, both of which act on different bodies.

$$\mathbf{F}_{12} = -\mathbf{F}_{21}$$

- The importance of Newton's second law of motion lies in that it gives the origin of force as:
If a particle experiences a force of magnitude F , it is because it exerts another force of the same magnitude F but in the opposite direction on some other particle.

2 Inertial frames of reference

Insert section content.

3 Non-inertial frames of reference

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